TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	69.0 ± 2.16
Correlation Stability	70.0 ± 17.20
Distribution Consistency	79.0 ± 6.53
Entropy Preservation	83.7 ± 3.09
Mean Z Score Drift	0.4 ± 0.09
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	20.0 ± 5.66
Data Density	100.0 ± 0.00
Coverage Score	78.0 ± 7.07
Feature Accuracy	83.7 ± 0.47
Novelty Score	0.7 ± 0.05

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Dataset Audit: parkinsons.csv

1. Dataset Portrait

Attribute	Value
Rows	195
Columns	23
Numeric features	23
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o
Folds (*13)	3
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
MDVP:Fo	154.229	145.600	-8.628	5.6%	OK
MDVP:Fhi	197.105	159.288	-37.817	19.2%	Warn
MDVP:Flo	116.325	129.358	+13.033	11.2%	Warn
MDVP:Jitter	0.006	0.005	-0.002	24.4%	Warn
MDVP:Jitter.1	0.006	0.005	-0.002	24.4%	Warn
MDVP:RAP	0.003	0.003	-0.001	23.2%	Warn
MDVP:PPQ	0.003	0.003	-0.001	19.3%	Warn
Jitter:DDP	0.010	0.007	-0.003	28.9%	Drift
MDVP:Shimmer	0.030	0.025	-0.005	16.2%	Warn
MDVP:Shimmer.1	0.030	0.025	-0.005	16.2%	Warn
Shimmer:APQ3	0.016	0.013	-0.002	15.6%	Warn
Shimmer:APQ5	0.018	0.015	-0.003	15.8%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	69.0 ± 2.16
Correlation Stability (*2)	70.0 ± 17.20
Distribution Consistency (*3)	79.0 ± 6.53
Entropy Preservation (*4)	83.7 ± 3.09
Mean Z Score Drift (*5)	0.4 ± 0.09
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	20.0 ± 5.66

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	78.0 ± 7.07
Feature Accuracy (*10)	83.7 ± 0.47
Novelty Score (*11)	0.7 ± 0.05

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9743589743589743	0.6410256410256411	0.9230769230769231	-0.05128205128205126	
F1	0.9806435394670694	0.7210522302706824	0.9410613707848731	-0.03958216868219577	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8888888888888889	0.6837606837606837	0.8547008547008547	-0.03418803418803421	
F1	0.9168253968253968	0.7705586429842723	0.8839932712215322	-0.03283212560386476	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9487179487179486	0.6837606837606837	0.9487179487179486	0	
F1	0.9622806387512269	0.7536878216123499	0.9612122626828509	-0.00106837606837606046	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9572649572649571	0.7350427350427351	0.9145299145299145	-0.04273504273504269	
F1	0.9680423280423282	0.8099856577998589	0.9308225705741234	-0.03721975746820464	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7777777777777778	0.7606837606837606	0.7863247863247863	0.00854700854700853	
F1	0.8614267676767677	0.83364898989899	0.8624368686868688	0.001010101010100995	

Model: LR

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8803418803418803	0.717948717948718	0.8205128205128206	-0.05982905982905984	
F1	0.91232423490488	0.7588766872569206	0.8703669023206455	-0.04195733258423462	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8461538461538463	0.6068376068376069	0.7777777777777778	-0.0683760683760684	
F1	0.8918089248277927	0.6844774273345702	0.850852598704888	-0.040956326122904674	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.717948717948718	0.6239316239316239	0.6837606837606837	-0.03418803418803421	
F1	0.749906191369606	0.692483660130719	0.7170385713837367	-0.03286761998586936	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8205128205128206	0.5726495726495726	0.8205128205128204	-3.700743415417188e-17	
F1	0.8836509065991988	0.6838396278051451	0.8687319832147417	-0.014918923384457031	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9401709401709403	0.717948717948718	0.8547008547008547	-0.08547008547008546	
F1	0.9570703384745131	0.7721706864564007	0.8892434988179669	-0.06782683965654616	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	119.992	157.302	74.997	0.00784	0.00784	0.0037	0.00554	0.01109	0.04374	0.04374
2	122.4	148.65	113.819	0.00968	0.00968	0.00465	0.00696	0.01394	0.06134	0.06134
3	116.682	131.111	111.555	0.0105	0.0105	0.00544	0.00781	0.01633	0.05233	0.05233
4	116.676	137.871	111.366	0.00997	0.00997	0.00502	0.00698	0.01505	0.05492	0.05492
5	116.014	141.781	110.655	0.01284	0.01284	0.00655	0.00908	0.01966	0.06425	0.06425
6	120.552	131.162	113.787	0.00968	0.00968	0.00463	0.0075	0.01388	0.04701	0.04701
7	120.267	137.244	114.82	0.00333	0.00333	0.00155	0.00202	0.00466	0.01608	0.01608
8	107.332	113.84	104.315	0.0029	0.0029	0.00144	0.00182	0.00431	0.01567	0.01567
9	95.73	132.068	91.754	0.00551	0.00551	0.00293	0.00332	0.0088	0.02093	0.02093
10	95.056	120.103	91.226	0.00532	0.00532	0.00268	0.00332	0.00803	0.02838	0.02838
11	88.333	112.24	84.072	0.00505	0.00505	0.00254	0.0033	0.00763	0.02143	0.02143
12	91.904	115.871	86.292	0.0054	0.0054	0.00281	0.00336	0.00844	0.02752	0.02752
13	136.926	159.866	131.276	0.00293	0.00293	0.00118	0.00153	0.00355	0.01259	0.01259
14	139.173	179.139	76.556	0.0039	0.0039	0.00165	0.00208	0.00496	0.01642	0.01642
15	152.845	163.305	75.836	0.00294	0.00294	0.00121	0.00149	0.00364	0.01828	0.01828
16	142.167	217.455	83.159	0.00369	0.00369	0.00157	0.00203	0.00471	0.01503	0.01503
17	144.188	349.259	82.764	0.00544	0.00544	0.00211	0.00292	0.00632	0.02047	0.02047
18	168.778	232.181	75.603	0.00718	0.00718	0.00284	0.00387	0.00853	0.03327	0.03327
19	153.046	175.829	68.623	0.00742	0.00742	0.00364	0.00432	0.01092	0.05517	0.05517
20	156.405	189.398	142.822	0.00768	0.00768	0.00372	0.00399	0.01116	0.03995	0.03995

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	243.439	250.912	232.435	0.0021	0.0021	0.00109	0.00137	0.00327	0.01419	0.01419
2	115.38	123.109	108.634	0.00332	0.00332	0.0016	0.00199	0.0048	0.01503	0.01503
3	112.876	148.826	106.981	0.005	0.005	0.0027	0.00289	0.0081	0.03999	0.03999
4	153.88	172.86	78.128	0.0048	0.0048	0.00232	0.00267	0.00696	0.04137	0.04137
5	142.729	162.408	65.476	0.00831	0.00831	0.00469	0.00419	0.01407	0.03485	0.03485
6	118.747	123.723	109.836	0.00331	0.00331	0.00168	0.00171	0.00504	0.01043	0.01043
7	145.678	157.234	132.456	0.00412	0.00412	0.00234	0.00256	0.00702	0.02234	0.02234
8	123.456	135.789	110.234	0.00345	0.00345	0.00178	0.00201	0.00534	0.01823	0.01823
9	167.89	180.234	150.678	0.00567	0.00567	0.00312	0.00345	0.00934	0.03012	0.03012
10	189.012	200.345	170.123	0.00234	0.00234	0.00123	0.00145	0.00367	0.01234	0.01234
11	134.567	145.678	120.345	0.00456	0.00456	0.00245	0.00278	0.00734	0.02567	0.02567
12	156.789	170.123	140.567	0.00378	0.00378	0.00201	0.00234	0.00612	0.02012	0.02012
13	110.417	131.067	103.37	0.00784	0.00784	0.00352	0.00514	0.01056	0.03715	0.03715
14	208.083	253.792	91.802	0.00757	0.00757	0.00428	0.00428	0.01285	0.06725	0.06725
15	244.99	272.21	239.17	0.00451	0.00451	0.00279	0.00237	0.00837	0.01897	0.01897
16	228.969	239.541	113.201	0.00238	0.00238	0.00136	0.0014	0.00408	0.01745	0.01745

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	223.365	238.987	98.664	0.00264	0.00264	0.00154	0.00151	0.00461	0.01906	0.01906
2	177.876	192.921	168.013	0.00411	0.00411	0.00233	0.00241	0.007	0.02126	0.02126
3	107.332	113.84	104.315	0.0029	0.0029	0.00144	0.00182	0.00431	0.01567	0.01567
4	142.167	217.455	83.159	0.00369	0.00369	0.00157	0.00203	0.00471	0.01503	0.01503
5	109.86	126.358	104.437	0.00874	0.00874	0.00398	0.00539	0.01193	0.03209	0.03209
6	201.464	210.565	195.708	0.00198	0.00198	0.00105	0.00115	0.00314	0.01194	0.01194
7	135.456	145.678	120.345	0.00456	0.00456	0.00234	0.00267	0.00702	0.01845	0.01845
8	210.123	230.456	190.789	0.00234	0.00234	0.00123	0.00145	0.00435	0.01234	0.01234
9	178.901	198.234	160.567	0.00345	0.00345	0.00189	0.00212	0.00634	0.01678	0.01678
10	120.567	130.678	110.456	0.00567	0.00567	0.00312	0.00345	0.00934	0.02234	0.02234
11	200.789	220.901	180.678	0.00123	0.00123	0.00078	0.00089	0.00267	0.01012	0.01012
12	150.345	160.456	140.234	0.00456	0.00456	0.00234	0.00256	0.00702	0.01923	0.01923
13	245.51	262.09	231.848	0.00235	0.00235	0.00127	0.00148	0.0038	0.01608	0.01608
14	115.38	123.109	108.634	0.00332	0.00332	0.0016	0.00199	0.0048	0.01503	0.01503
15	95.056	120.103	91.226	0.00532	0.00532	0.00268	0.00332	0.00803	0.02838	0.02838
16	143.533	162.215	65.809	0.01101	0.01101	0.00647	0.00467	0.01941	0.05384	0.05384

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B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	118.747	123.723	109.836	0.00331	0.00331	0.00168	0.00171	0.00504	0.01043	0.01043
2	110.417	131.067	103.37	0.00784	0.00784	0.00352	0.00514	0.01056	0.03715	0.03715
3	156.405	189.398	142.822	0.00768	0.00768	0.00372	0.00399	0.01116	0.03995	0.03995
4	202.805	231.508	86.232	0.0037	0.0037	0.00189	0.00211	0.00568	0.01997	0.01997
5	128.001	138.052	122.08	0.00436	0.00436	0.00137	0.00166	0.00411	0.02297	0.02297
6	149.689	160.368	133.608	0.00257	0.00257	0.00116	0.00134	0.00349	0.01346	0.01346
7	134.567	145.678	120.345	0.00456	0.00456	0.00234	0.00278	0.00702	0.02567	0.02567
8	145.678	167.89	123.456	0.00567	0.00567	0.00312	0.00345	0.00936	0.03245	0.03245
9	156.789	178.901	130.567	0.00345	0.00345	0.00178	0.00212	0.00534	0.02012	0.02012
10	167.89	189.012	140.678	0.00678	0.00678	0.00345	0.00401	0.01023	0.0389	0.0389
11	178.901	200.123	150.789	0.00234	0.00234	0.00123	0.00156	0.00345	0.01234	0.01234
12	189.012	210.234	160.89	0.00789	0.00789	0.00456	0.00478	0.01345	0.04567	0.04567
13	242.852	255.034	227.911	0.00225	0.00225	0.00117	0.00139	0.0035	0.01494	0.01494
14	139.224	586.567	66.157	0.03011	0.03011	0.01854	0.01628	0.05563	0.09419	0.09419
15	186.163	197.724	177.584	0.00298	0.00298	0.00165	0.00175	0.00496	0.01495	0.01495
16	116.879	131.897	108.153	0.00788	0.00788	0.00334	0.00493	0.01003	0.02645	0.02645